

CLÉMENT ROMAC

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Age : 29 ♦ <https://clementromac.github.io/>

EDUCATION

University of Bordeaux, Bordeaux Ph.D. in Computer Science Pierre-Yves Oudeyer (Inria, FLOWERS) & Thomas Wolf (Hugging Face)	<i>2022 - 2026</i>
University of Bordeaux, Bordeaux MSc. in Computer Science Algorithms and Models specialisation	<i>2019 - 2020</i>
Ynov Informatique Ingésup, Bordeaux Degree in Computer Science engineering Data Science specialisation	<i>2014 - 2019</i> Rank : 1
High School Les Iris, Lormont High School Degree, with the highest honours	<i>2011 - 2014</i> Overall Notation : 17.75/20

WORK EXPERIENCE

Hugging Face <i>Research Scientist</i> Studying how to turn LLMs into autonomous RL agents.	<i>January 2022 - January 2026</i>
Inria (FLOWERS team), Bordeaux <i>PhD student</i> Studying how to turn LLMs into autonomous RL agents.	<i>January 2022 - January 2026</i>
Inria (FLOWERS team), Bordeaux <i>Research Engineer</i> Working on an experimental platform to apply curiosity-driven algorithms to automatic discovery of Complex Systems (e.g., self-organizing)	<i>September 2020 - December 2021</i>
Inria (FLOWERS team), Bordeaux <i>Research intern</i> - Worked on TeachMyAgent, a benchmark platform for Automatic Curriculum Learning applied to Deep Reinforcement Learning algorithms : - Created procedurally generated Box2D environments and embodiments - Implemented state-of-the-art Automatic Curriculum Learning methods - Built a benchmark setup and benchmarked the methods - Paper accepted at ICML 21'	<i>March 2020 - August 2020</i>
Weenove, Bordeaux <i>Part time Data Scientist</i> - In charge of the Machine Learning R&D : - Implemented an Automated Machine Learning service - Made this service available (Beta) in a Business Intelligence Software (Biwee) - Lead the Machine Learning R&D projects	<i>October 2017 - February 2020</i>

— Data Science projects

Scalian, Bordeaux

Summer 2017

Intern Data Scientist

- Data Science projects :
 - Time Series Anomaly Detection
 - Barcode localization on drone pictures with Deep Learning
 - Data Exploration project

Weenove, Bordeaux

June 2016 - May 2017

Part time Intern Data Scientist

- Machine Learning R&D :
 - Implemented an Automated Machine Learning service
 - Made this service available (Beta) in a Business Intelligence Software (Biwee)

PROJECTS

ChatGPT expliqué en 5 minutes (2023)

A series of short videos explaining Large Language Models. The project can be found **here**.

Lamorel (2022)

Lamorel is a Python library designed for RL practitioners eager to use LLMs. It relies on a distributed client-server architecture, which allows seamless scaling (**repository**).

Automated Discovery Tool (2021)

A software for assisted and automated discovery of patterns in the exploration of complex systems (**repository**).

Interactive Deep RL Demo (2021)

I supervised the project of turning the environments I developed for TeachMyAgent into an on-browser interactive demo of Deep Reinforcement Learning agents' adaptation to unknown 2D tasks. The project can be found **here**.

PUBLICATIONS

MAGELLAN : Metacognitive predictions of learning progress guide autotelic LLM agents in large goal spaces

Loris Gaven, Thomas Carta, Clément Romac, Sylvain Lamprier, Olivier Sigaud, Pierre-Yves Oudeyer. *Forty-Second International Conference on Machine Learning (ICML) 2025*.

WorldLLM : Improving LLMs' world modeling using curiosity-driven theory-making

Guillaume Levy, Cedric Colas, Pierre-Yves Oudeyer, Thomas Carta, Clement Romac. *Multi-disciplinary Conference on Reinforcement Learning and Decision Making (RLDM) 2025*.

Reinforcement Learning for Aligning Large Language Models Agents with Interactive Environments : Quantifying and Mitigating Prompt Overfitting

Mohamed Salim Aissi, Clement Romac, Thomas Carta, Sylvain Lamprier, Pierre-Yves Oudeyer, Olivier Sigaud, Laure Soulier, Nicolas Thome. *Findings of the Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL) 2025*.

SAC-GLAM : Improving Online RL for LLM agents with Soft Actor-Critic and Hindsight Relabeling

Loris Gaven, Clément Romac, Thomas Carta, Sylvain Lamprier, Olivier Sigaud, Pierre-Yves Oudeyer. *Intrinsically Motivated Open-ended Learning Workshop, NeurIPS 2024*.

Grounding Large Language Models in Interactive Environments with Online Reinforcement Learning

Clément Romac*, Thomas Carta*, Thomas Wolf, Sylvain Lamprier, Olivier Sigaud, Pierre-Yves Oudeyer. *Fortieth International Conference on Machine Learning (ICML) 2023*. * equal contribution.

TeachMyAgent : a Benchmark for Automatic Curriculum Learning in Deep RL

Clément Romac* and Rémy Portelas* and Katja Hofmann and Pierre-Yves Oudeyer. *Thirty-eighth International Conference on Machine Learning (ICML) 2021*. * equal contribution.

Meta Automatic Curriculum Learning

Rémy Portelas and Clément Romac and Katja Hofmann and Pierre-Yves Oudeyer. 2020 (Preprint). abs/2011.08463.

Partially Observable Markov Decision Process with Minecraft

Romac et Béraud. 2019 (Preprint). abs/1903.04311.

LANGUAGES

French	Native speaker
English	Advanced (TOEIC : 980/990, 2019)

TEACHING

Ecole Nationale Supérieure de Cognitique Bordeaux <i>Last year engineering students from AI Specialisation</i> <i>Transformers and LLMs - 14h (Lectures and seminars)</i>	<i>October 2025</i>
Ecole Nationale Supérieure de Cognitique Bordeaux <i>Last year engineering students from AI Specialisation</i> <i>Transformers and LLMs - 14h (Lectures and seminars)</i> <i>Reinforcement Learning - 7h (Lectures and seminars)</i>	<i>October 2024</i>
Ecole Nationale Supérieure de Cognitique Bordeaux <i>Last year engineering students from AI Specialisation</i> <i>Transformers and LLMs - 7h (Lectures and seminars)</i>	<i>October 2023</i>
Ynov Informatique Ingésup Bordeaux <i>Masters student, Data Science Specialisation</i> <i>Machine Learning Major - 90h (Lectures and seminars)</i>	<i>November 2018 - December 2018</i>

STUDENTS SUPERVISED

Ryan Boldi <i>Undergrad internship</i>	<i>May-September 2025</i>
Loris Gaven <i>Master's internship</i>	<i>May-September 2024</i>
Guillaume Levy <i>Master's internship</i>	<i>May-September 2024</i>
Matthieu Perie <i>Part-time master's student</i>	<i>September 2021 - September 2023</i>
Paul Germon <i>Master's internship</i>	<i>May-September 2021</i>

SERVICE AND LEADERSHIP

Co-founded the AI working group of Ynov Informatique Ingésup Bordeaux

— We gather every two weeks on Wednesdays to work on AI projects.

Co-founded and animated (until 2019) the meetup “Les nuits des réseaux de neurones”

— Talks and workshops on Deep Learning starting in the evening and continuing throughout the night.

— Held every month or two months, open to everyone (270 members).

— <https://www.meetup.com/fr-FR/Les-nuits-des-reseaux-de-neurones/>

Co-organizer of the meetup “Bordeaux Machine Learning Meetup”

— Talk and workshops once a month (+1400 members)

— <https://www.meetup.com/fr-FR/Bordeaux-Machine-Learning-Meetup/>

I have also been involved in various dissemination initiatives, in particular towards education (I am co-leading the development of the “ChatGPT explained in 5 minutes” series of videos that is used in French classrooms and now embedded in official trainings for workers from all French ministries)

HOBBIES

Sport : In my spare time, I enjoy doing sport, from outdoor activities (hiking, mountaineering, running, trail, canoeing...) to football, tennis padel, badminton...

Music : I have been playing bass guitar for several years, including in a jazz fusion band.